TAITIEN
Our Performance Your Reputation

100MHz *Low Noise/Low G-Sensitivity* OCXO
NA-100M-6800 series

NA-100M-6800 Series in 25.4x25.4mm DIP package



NA-100M-6800 series is a 100.000 MHz high performance (VC)OCXO offering low phase noise(LPN), low G sensitivity(LGS) and tight frequency stability down to ±50ppb(-20ºC to +70ºC). The part comes in a small hermetically sealed through hole package which makes it suitable for humid environmental conditions

RoHS Compliant Standard

FEATURES

- ***Low Phase Noise & Low G-Sensitivity***
- Small Hermetically Sealed Package
- Tight Frequency Stability
- Low Power Consumption
- Fast Warm-up Time
- Electrical Frequency Tuning Input
- Reference Voltage Output
- RoHS-Compliant (lead-free)

APPLICATIONS

- Instrument Reference
- Microwave Communication
- Clock Reference for Microwave Signal Source
- Synthesizer Reference Clock
- Test & Measurement
- Telecom Systems
- Radar Systems

ELECTRICAL SPECIFICATIONS

Test conditions: VDC = +12 V; VCO = +5 V; at +25 ± 3°C unless otherwise identified

1. OUTPUT (PIN = "R.F. OUTPUT")

	Parameter	Min.	Typ.	Max.	Unit	Test Condition
1.1.	Frequency (Fo)		100.000000		MHz	
1.2.	Initial Accuracy	-0.3		+0.3	ppm	@ +25 ±1°C
						after turn on power 30 minutes
						Vco=+5V
1.3.	Waveform		Sine wave			
1.4.	Level	+10			dBm	
1.5.	Load		50		Ω	
1.6.	Harmonics			-30	dBc	
1.7.	Spurious			-80	dBc	10Hz ~ 1KHz from carrier
				-100	dBc	1KHz ~ 1MHz from carrier

2. FREQUENCY STABILITY

	Parameter	Min.	Typ.	Max.	Unit	Test Condition	
2.1.	Ambient	±20, ±50, ±100, ±200			ppb	referred to 25°C	**Refer to Table 1 : Ordering Information**
		-20°C ~ +70°C -40°C ~ +85°C			°C		
2.2.	Aging						
	Daily	-5		+5	ppb	after 30 days	
	Yearly	-500		+500	ppb		
	10 Years	-2		+2	ppm		
2.3.	Voltage	-5		+5	ppb	±5% change	
2.4.	Short term			0.02	ppb	root Allan variance for τ=1 sec	
2.5.	Load	-5		+5	ppb	±5% change	
2.6.	Warm-up	-50		+50	ppb	in 5 minutes @ +25 ±1°C	referred to 1 hour
2.7.	G-Sensitivity (each axis)			1	ppb/g		

2.8. Phase Noise (Max.)	Option A	Option B	Option C	Option D	Option E		Refer to Table 1 : Ordering Information
	-93	-97	-100	-105	-110	dBc/Hz	@ 10Hz
	-125	-130	-135	-138	-140	dBc/Hz	@ 100Hz
	-157	-160	-162	-164	-170	dBc/Hz	@ 1KHz
	-173	-173	-174	-175	-176	dBc/Hz	@ 10KHz
	-177	-177	-177	-177	-177	dBc/Hz	@ 100KHz
	-178	-178	-177	-177	-178	dBc/Hz	@ 1MHz

3. ELECTRICAL FREQUENCY ADJUSTMENT (PIN = "VCO INPUT")

	Parameter	Min.	Typ.	Max.	Unit	Test Condition
3.1.	Tuning Range	±3			ppm	Referenced to frequency at nominal Center Voltage. Phase Noise Option A ~ D
		±2.5				Referenced to frequency at nominal Center Voltage. Phase Noise Option E
3.2.	Control Voltage	0		+10.0	V	
3.3.	Slope		Positive			
3.4.	Center Voltage		+5		V	
3.5.	Linearity	-10		+10	%	

4. INPUT POWER (PIN = "+VDC")

	Parameter	Min.	Typ.	Max.	Unit	Test Condition
4.1.	Voltage	+11.4	+12	+12.6	V	
4.2.	Current					
	Steady State			150	mA	Phase Noise Option A ~ D
				167		Phase Noise Option E
	During Warm-Up			350	mA	Phase Noise Option A ~ D
				400		Phase Noise Option E

5. REFERENCE VOLTAGE (PIN = "REFERENCE VOLTAGE")

	Parameter	Min.	Typ.	Max.	Units	Test Condition
5.1.	Voltage	+9.5	+10	+10.5	V	

6. ENVIRONMENTAL

	Parameter	Reference Std.	Test Condition
6.1.	Operable Temperature	-40°C to +85°C	Note 1
6.2.	Storage Temperature	-45°C to +90°C	
6.3.	Humidity	MIL-STD-202, Method 103 Test Condition A	95% RH @ +40°C,non-condensing,240 hours
6.4.	Vibration (non-operating)	MIL-STD-202, Method 201	0.06" Total p-p, 10 to 55 Hz
6.5.	Shock (non-operating)	MIL-STD-202, Method 213, Test Condition J	30g, 11ms, half-sine

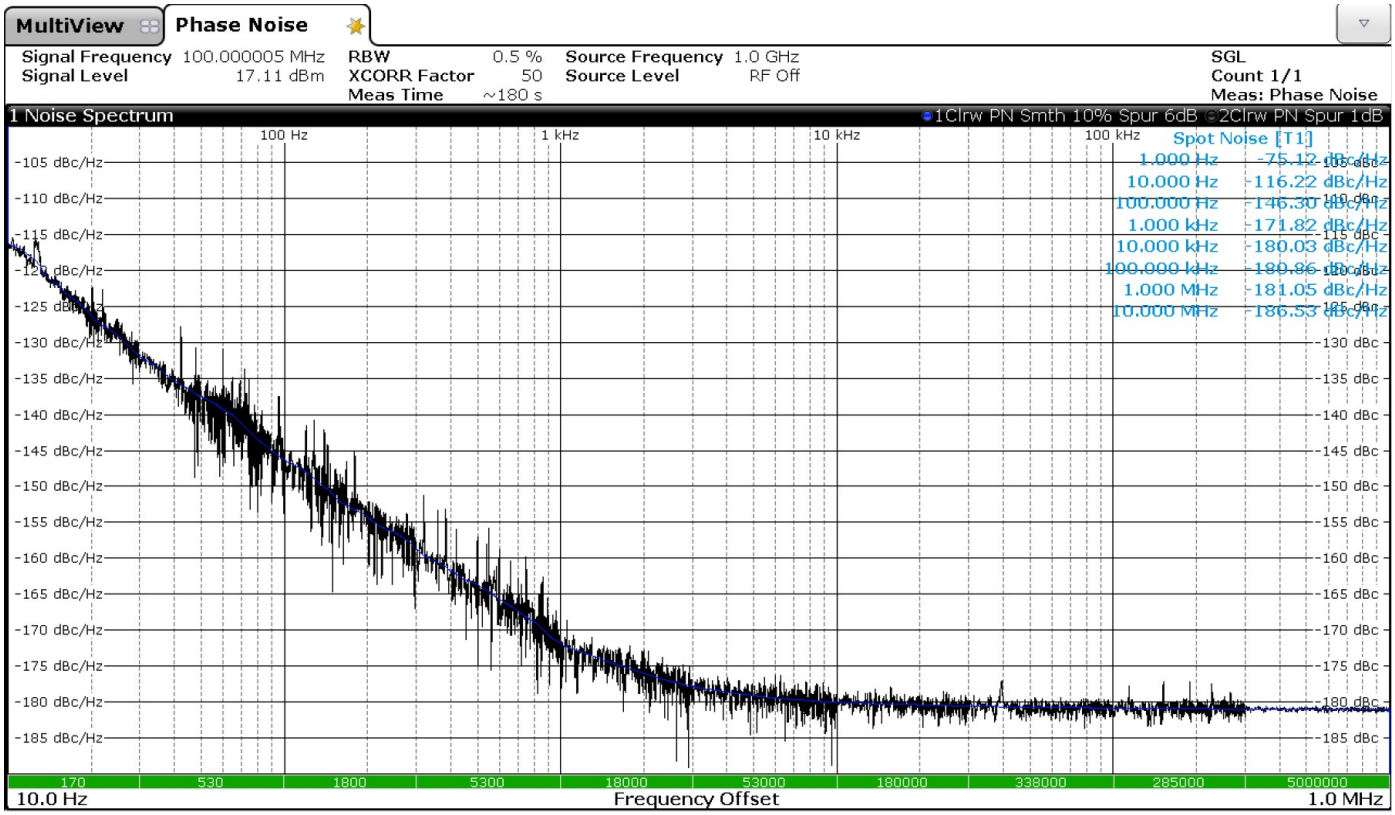
Note 1 : Output maintained over this temperature range. Other requirements of this specification may not be met when operating outside the temperature range in 2.1.

Table 1 : ORDERING INFORMATION

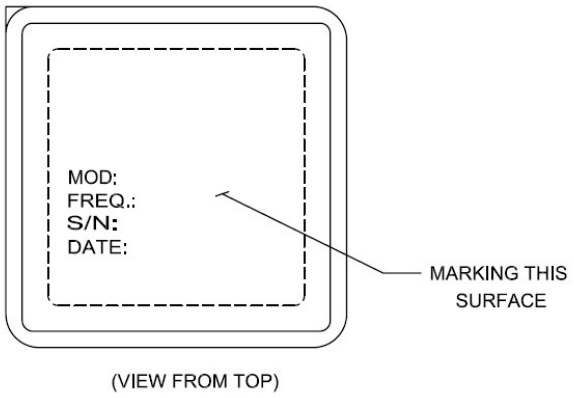
Ambient Temp. (°C) / Option		Phase Noise Option				
		A	B	C	D	E
-20°C ~+70°C	±100 ppb	NA-100M-6800	NA-100M-6801	NA-100M-6802	NA-100M-6803	NA-100M-6804
	±50 ppb	NA-100M-6810	NA-100M-6811	NA-100M-6812	NA-100M-6813	NA-100M-6814
	±20 ppb	NA-100M-6830	NA-100M-6831	NA-100M-6832	NA-100M-6833	NA-100M-6834
-40°C~+85°C	±200 ppb	NA-100M-6860	NA-100M-6861	NA-100M-6862	NA-100M-6863	NA-100M-6864
	±100 ppb	NA-100M-6820	NA-100M-6821	NA-100M-6822	NA-100M-6823	NA-100M-6824
	±50 ppb	NA-100M-6870	NA-100M-6871	NA-100M-6872	NA-100M-6873	NA-100M-6874

Other specifications may be available upon request.

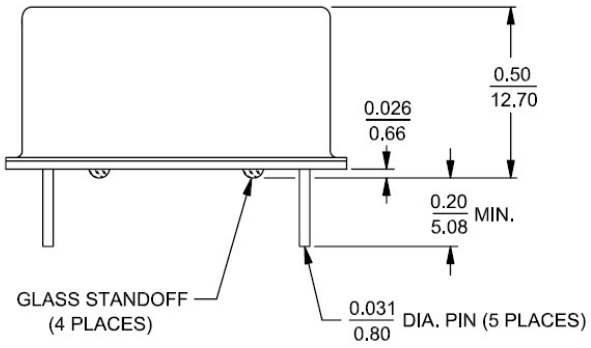
Phase Noise Test Data



OUTLINE DRAWING



PIN CONNECTIONS	
PIN	FUNCTION
1	R. F. OUTPUT
2	0 VOLTS & CASE
3 (See Note 1)	VCO INPUT or NOT CONNECTED
4 (See Note 1)	REFERENCE VOLTAGE or NOT CONNECTED
5	+VDC

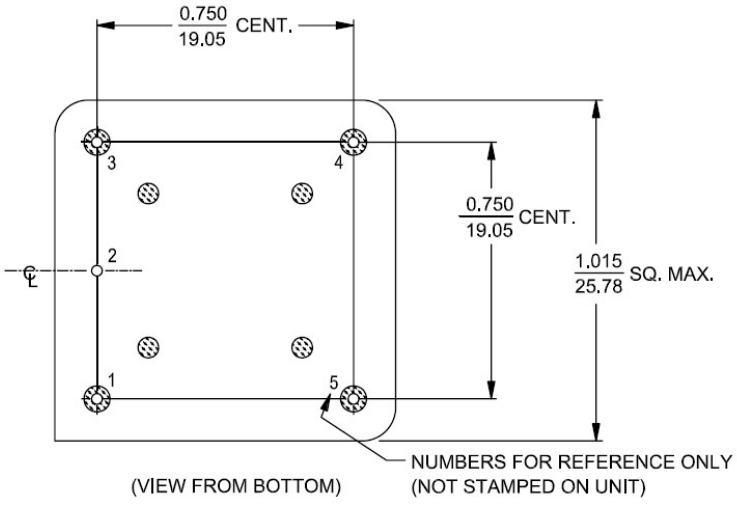


Note 1. If the specification does not specify parameters for either PIN3 or PIN4 then that respective PIN is NOT internally CONNECTED.

MARKING







INCH / mm (REFERENCE ONLY)

REV.	DATE	DESCRIPTION	ECO,NO.	DRAWN	DESIGNED	CHECKED	APPROVED	SCALE
1.0	10/29/2015		N00????	HCY	SY	SY	IMYeh	2:1
2.0	05/08/2017	Pin3 Add NOT CONNECTED Option,Pin4 Add REFERENCE VOLTAGE Option	M00????	HCY	SY	CL	IMYeh	DATE
								05/08/2017

TAITIEN ELECTRONICS(NANJING)CO.,LTD.

TOLERANCES:
UNLESS OTHERWISE SPECIFIED:
ANGLES: ±1 DEGREE
FRACTIONS:±1/32 INCH
DECIMALS: .XX±.015, .XXX±.010 INCH

M-NA-005-N

REV. 2.0